Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	High-Performance Solid-State Polymer Electrolytes for Lithium–Sulfur Batteries			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		Electrical and Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Batteries	Battery Materials
Concept of Proposed Technology	This project aims to develop high-performance solid-state (or quasi-solid-state) polymer electrolytes for lithium–sulfur (LiS) batteries, with the U.S. research team leading electrolyte material innovation and the Korean industrial partner demonstrating prototype LiS cells. Our advanced polymer electrolytes enable a desirable combination of properties, including high ionic conductivity with minimal liquid electrolyte content, high lithium-ion selectivity over polysulfides, self-healing capability, strong electrode adhesion, and excellent electrochemical and mechanical stability, outperforming conventional ceramic and polymer electrolytes. By addressing key challenges such as polysulfide shuttling, interfacial instability, and safety limitations, this joint R&D effort will accelerate the development of safe, high-energy-density LiS batteries for next-generation energy storage applications.			

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Rationale for Joint R&D Project	Lithium–sulfur (LiS) batteries are a key next-generation secondary battery technology due to their high energy density and use of abundant materials, but commercialization is limited by polysulfide shuttling, interfacial instability, and safety challenges. This joint R&D project addresses these barriers through advanced solid-state (or quasi-solid-state) polymer electrolytes. The U.S. research team provides high-risk, high-impact electrolyte material innovation and electrochemical characterization, while the Korean industrial partner leads LiS cell assembly and prototype validation. This collaboration accelerates development of safe, high-performance battery materials and cells, directly supporting Korea's national strategic secondary battery technology and industrial competitiveness.
Key Words	1. Lithium-Sulfur Batteries 2. Next-Generation Secondary Batteries 3. High-Energy-Density Energy Storage 4. Solid-State Polymer Electrolytes 5. Safe and Durable Battery Materials

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

A. Overarching Goal: A Versatile Polymer Platform for Next-Gen Li-S Batteries

The main objective of this R&D project is to develop advanced polymeric electrolyte materials that enable high-performance lithium–sulfur batteries (LSBs) across multiple electrolyte architectures, including quasi-solid-state, hybrid, multilayer, and composite systems. Rather than being constrained to a single battery configuration, this project centers on polymer material development as a core enabling technology. This approach provides a versatile and scalable platform capable of integrating with limited liquid electrolyte and/or dispersed inorganic lithium-ion conductors. Ultimately, the goal is to deliver transferable, scalable polymer materials that form a foundational technology for next-generation LSBs, ensuring improved safety, durability, and energy density while maintaining compatibility with industrial processing.

B. Specific Technical Aims: Multi-Functional Polymer Electrolyte Matrix Design

This project aims to design and optimize functional polymer matrices, building upon our successful demonstration of multifunctional electrochemical membranes in LSBs, specifically BPSA and zPU [1, 2]. These polymer electrolytes are designed to deliver high lithium-ion (Li^+) conductivity, suppressed polysulfide (PS) crossover (shuttle mechanism), strong interfacial adhesion, and excellent electrochemical and mechanical stability, resulting in enhanced cycling stability (**Fig. 1**).

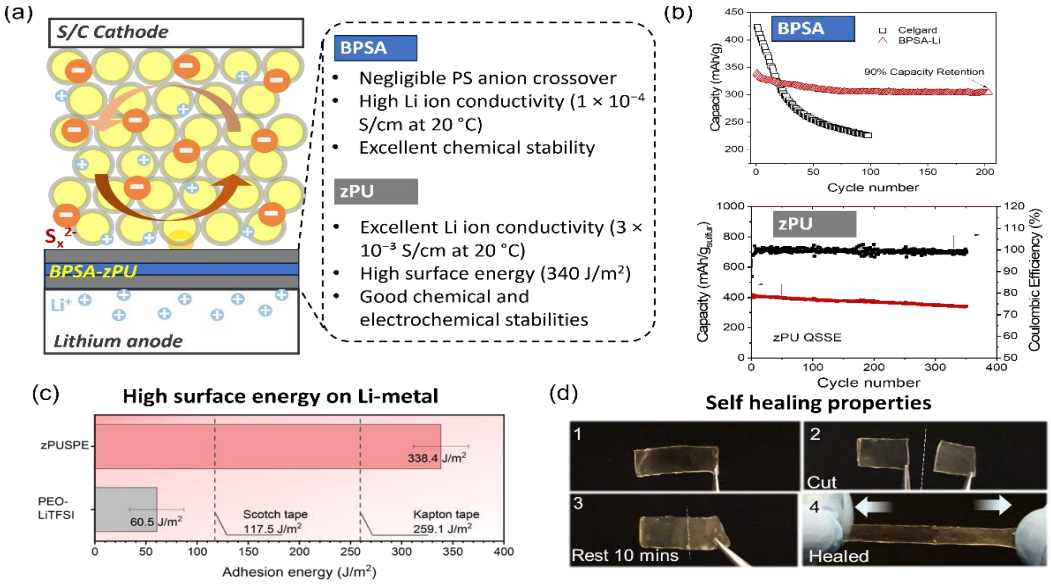


Figure 1. (a) Quasi-solid state electrolyte system (Multilayered electrolyte) comprising Li-conducting polymer BPSA sandwiched between zwitterionic polyurethane (zPU). (b) Discharge capacity of BPSA and Celgard at 0.75C and room temperature; Discharge capacity and coulombic efficiency versus cycle number measured for zPU QSSE. (c) Surface adhesion energy of zPU with lithium metal in comparison to traditional PEO-SPE, Scotch tape, and Kapton tape. (d) Photography of the seal-healing process.

The developed polymeric materials will be engineered as a versatile materials platform capable of supporting multiple electrolyte architectures for LSBs. Specifically, the polymer matrices will be designed to serve the following three functions (**Fig. 2**):

(i) Quasi-solid-state electrolytes (QSSE): Polymer matrices that function as robust QSSEs capable of achieving high electrochemical performance with low liquid electrolyte content, while maintaining sufficient ionic conductivity, mechanical integrity, and interfacial stability with lithium metal anodes and sulfur cathodes.

(ii) Composite host matrices: Polymeric host materials that enable the effective incorporation and uniform dispersion of inorganic Li^+ conductors (e.g., LATP) within the polymer matrix. These

composite electrolytes are intended to enhance Li^+ conductivity and selectivity while preserving the processability, mechanical compliance, and interfacial contact advantages of polymers.

(iii) Multi-layer electrolyte: Engineered multi-layer polymer electrolyte structures that combine layers with distinct mechanical and chemical functionalities, such as a mechanically robust core layer and interfacially compliant surface layers, to balance high Li^+ selectivity (including effective suppression of polysulfide shuttling), strong adhesion to lithium metal anodes, and stable contact with sulfur cathodes. For example, a tri-layer electrolyte system can be constructed with an advanced biphenyl-based single Li^+-conducting BPSA layer centrally positioned and effectively encapsulated between zPU layers. This architecture achieves a desirable combination of properties, including high ionic conductivity with minimal liquid electrolyte content (<25 wt%), self-healing capability, strong electrode adhesion, and excellent electrochemical and mechanical stability, thereby outperforming conventional ceramic and polymer electrolytes.

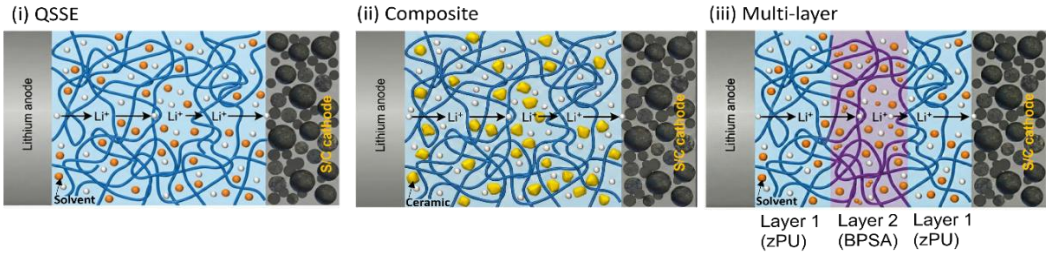


Figure 2. Schematic illustration of three distinct polymer electrolyte configurations. (i) Quasi-solid-state electrolyte (QSSE) depicting a uniform, homogeneous polymer matrix. (ii) Composite host matrix showing inorganic solid conductors (irregular yellow particles) dispersed within the polymer phase. (iii) Multi-layer electrolyte featuring a distinct engineered trilayer structure with a central core layer (single-ion conducting BPSA) sandwiched between two outer surface layers (zPU).

C. Strategic Objectives: Overcoming Ceramic and Polymeric Electrolyte Limitations

The central technical objective of this project is to overcome the fundamental limitations of both inorganic solid-state electrolytes and conventional solid polymer electrolytes in LSB systems.

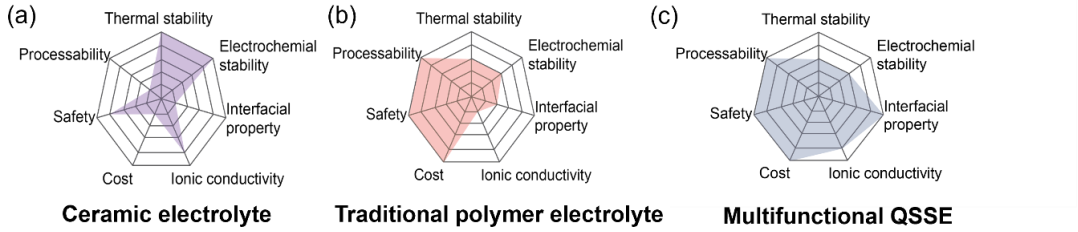
Inorganic solid electrolytes are attractive due to their high Li^+ conductivity, strong cation selectivity, and excellent electrochemical and thermal stability; however, their practical use in LSBs is severely constrained by inherent brittleness, mechanical fragility, and poor interfacial compatibility with electrodes (**Fig. 3a**). These challenges are exacerbated by the large volume changes and heterogeneous interfaces associated with sulfur cathodes and Li metal anodes, often resulting in high interfacial resistance, mechanical failure, and poor cycling stability.

Solid polymer electrolytes (SPEs), by contrast, offer mechanical flexibility, elasticity, improved electrode contact, and ease of processing (**Fig. 3b**). Nevertheless, conventional SPEs suffer from low room-temperature Li^+ conductivity and selectivity, with typical Li^+ transference numbers (t^+) below 0.4. Despite extensive efforts (e.g., the incorporation of fillers, plasticizers, copolymerization, and polymer blending) simultaneously achieving high conductivity, selectivity, and mechanical robustness in SPEs remains a major challenge. These challenges are further amplified in LSBs, where polysulfide dissolution and migration, combined with dynamic electrode volume changes, impose stringent and often conflicting requirements on electrolyte materials.

Our multifunctional polymer electrolyte strategy can directly address these coupled limitations through rational polymer chemistry and microstructural design that moves beyond traditional SPE paradigms (**Fig. 3c**) [2]. By tailoring polymer backbones, ion-coordination environments, and segmental dynamics, the developed polymer matrices aim to enable decoupled Li^+ transport, enhanced selectivity, controlled polysulfide transport, and robust interfacial stability. In composite and hybrid electrolyte architectures, these polymer matrices further function as active ionic and mechanical frameworks, enabling uniform dispersion of inorganic conductors and low polymer–ceramic interfacial resistance. Through this integrated approach, advanced polymer electrolytes are

R&D Target Specifications

established as a strategic materials platform for scalable, durable, and high-performance LSBs across quasi-solid-state, composite, and multilayer configurations.



Ceramic electrolyte **Traditional polymer electrolyte** **Multifunctional QSSE**

Figure 3. Performance radar maps of (a) ceramic electrolyte, (b) polymer electrolyte, and (c) our multifunctional QSSE system [2].

A. Quantitative Performance Targets (KPIs)

The R&D target specifications are defined to ensure that the developed polymeric electrolyte materials function as a core, transferable technology platform for LSBs, enabling quasi-solid-state, hybrid, composite, and multilayer electrolyte systems with improved performance, safety, and manufacturability. We have established the following key performance indicators (KPIs) to ensure the developed quasi-solid-state LSB exceeds the limitations of current liquid/polymeric/ceramic-based systems.

Category	**Key Performance Indicator (KPI)**	**Current Standard (Liquid/Conventional Li-S)** [3, 4]	**Project Final Target**	**Measurement Condition**
Material (Polymer electrolyte)	Ionic Conductivity, S/cm	< 10^{-4} (PEO-based SPE at 25 °C)	≥1.0×10^{-3}	@ 25°C, EIS Method
	Li^+ Transference Number (t^+)	<0.4	≥0.9	Bruce-Vincent Method
	Electrochemical Stability, V	<4.0 V (typical liquid Li–S systems)	≥4.5 V (no major current rise)	vs. Li/Li^+, LSV Scan
	Polysulfide crossover, %	High (100%)	≤5% (≥ 95% reduction)	Permeation cell, UV-Vis quantification
Cell Performance	Gravimetric Energy Density, Wh/kg	~300-350	≥500	Coin or pouch Cell (0.3C)
	Cycle Stability/capacity retention	100-200 cycles, < 70% reduction	≥500 and ≥ 80% retention	Pouch or coin cell (0.3C), ≥5 $mg\cdot cm^{-2}$ S, lean E/S
	Coulombic efficiency, %	~95–98% (typical Li–S with shuttle)	≥ 99.5% over steady-state cycling	Same cells as above, exclude initial formation cycles
	Electrolyte/Sulfur (E/S) Ratio, μL/mg	>5.0 (Flooded)	≤3.0	Defined electrolyte volume / S mass
	Sulfur Loading, mg/cm^2	2.0~3.0	≥5.0	Cathode areal sulfur loading
Safety	Flammability	Highly Flammable	Meets UL 94 V-0	UL 94 Vertical Burn

B. Detailed Technical Specifications

1) Polymeric Electrolyte Material Targets

- High Conductivity & Selectivity: Achieve ionic conductivity ≥ 2.0×10^{-3} S/cm at 25–40 °C with target $t^+ \geq 0.9$ in Li^+-selective domains.
- Polysulfide Shuttle Suppression: Reduce polysulfide permeability by ≥95% relative to conventional liquid electrolyte/Celgard, enabling stable Coulombic efficiency ≥99.0%.
- Mechanical Robustness
  - Elastic Modulus: Tunable soft-solid regime, approx. 50–500 MPa at 25 °C, to accommodate sulfur

	volume change and enable self-healing and strong Li adhesion, consistent with our prior zPU-SPE designs [2]. 0.7–1.5 GPa to provide dimensional stability and polysulfide-blocking, consistent with our prior BPSA designs [1]. - Self-Healing: Self-healing ≥80% recovery of ionic continuity after mechanical damage based on our prior work [2]. 2) Composite & Multilayer Architecture Targets ▪ Composite Uniformity: Achieve inorganic conductor loading of ~40 wt% within polymer matrix without phase separation or agglomeration. ▪ Interfacial Resistance: Limit the polymer–ceramic interfacial resistance increase to ≤30–50% relative to the pristine polymer, ensuring efficient ion transfer across boundaries. ▪ Layer Adhesion: In multilayer systems, demonstrate zero delamination after >500 cycles. 3) Scalability & Manufacturability (Korean SME Lead) ▪ Fabrication Method: Fully compatible with Roll-to-Roll (R2R) Slot-Die Coating. ▪ Thickness Control: Target electrolyte thickness of ≤50 μm to maximize volumetric energy density. ▪ Prototype Compatibility: The electrolyte films must be coatable, dried, and laminated using the Korean SME's existing R2R and pouch assembly tools (10 Ah-class), without new capital equipment beyond minor tooling changes.
R&D Plan	The project will follow a systematic "Material-to-Device" development workflow. It begins with polymer electrolyte synthesis and physicochemical/electrochemical characterization (Phase 1), then transitions to laboratory-scale LSB cell demonstrations using three electrolyte implementations (QSSE, composite, layered) (Phase 2). The final phase establishes scale-up readiness, including polymer scale synthesis and R2R–compatible manufacturing of electrolyte films and multilayer structures for prototype translation (Phase 3). **Phase 1: Polymeric Electrolyte Materials Synthesis and Characterization** **Task 1. Polymer synthesis and formulation** • Synthesize functional polymer backbones and networks • Formulate QSSE-ready compositions (controlled liquid content), composite-ready matrices (filler compatibility), and lamination-ready layers (adhesion/compatibility) **Task 2. Physicochemical characterization** • Chemical structure confirmation (e.g., FTIR/NMR as applicable), thermal properties (DSC/TGA), morphology and phase behavior (e.g., SEM/AFM/SAXS as available), mechanical properties (DMA/tensile) and self-healing metrics. **Task 3. Electrochemical characterization (materials level)** • Ionic conductivity vs temperature (EIS), lithium transference number (t^+), Electrochemical stability window (LSV), polysulfide crossover/permeability tests (H-cell or equivalent), and Li|electrolyte interfacial impedance evolution (symmetric Li cells). **Phase 2: Architecture Demonstration for LSB Cells at Laboratory Scale** **Task 4. QSSE demonstration** • Prepare QSSE films/matrices with low liquid content and stable interfaces • Validate Li|QSSE stability and polysulfide suppression behavior **Task 5. Composite electrolyte demonstration** • Incorporate inorganic Li^+ conductors (target 20–40 wt%) • Optimize dispersion, polymer–filler interfaces, and mechanical integrity • Quantify polymer/inorganic interfacial resistance and its impact on cell impedance **Task 6. Layered electrolyte demonstration** • Fabricate multilayer structures (e.g., tough core + compliant/adhesive skins) • Evaluate layer adhesion, delamination resistance, and selective transport functionality **Task 7. Lab-scale LSB cell assembly and testing** • Assemble coin/pouch-prototype laboratory cells using each architecture

	• Test under practical conditions (lean electrolyte targets, meaningful sulfur loading) • Diagnose limiting mechanisms (shuttle, interfacial degradation, cathode utilization, mechanical failure) **Phase 3: Polymer Scale-Up and R2R Manufacturability Demonstration** **Task 8. Polymer scale synthesis and batch-to-batch reproducibility** • Scale polymer synthesis (multi-kg or pilot-lab scale as appropriate) • Establish QC metrics: molecular weight/viscosity windows, functional group density, thermal/mechanical property acceptance ranges **Task 9. R2R–compatible film manufacturing** • Demonstrate continuous casting/coating of electrolyte films (target thickness ≤50–100 μm) • Demonstrate R2R lamination for layered electrolyte architectures • Evaluate uniformity, defect tolerance, and throughput-relevant parameters (drying, solvent handling, adhesion) **Task 10. Scale-relevant validation** • Confirm that scaled and R2R–made materials retain key properties (e.g., conductivity, t^+, stability window, polysulfide control, interfacial impedance stability) • Assemble and test lab-scale LSB cells with R2R–fabricated electrolyte films to confirm performance retention **Timeline Summary** • **Months 1–12:** Polymer synthesis + physicochemical & electrochemical characterization; down-select candidates • **Months 10–24:** Demonstrate QSSE, composite, layered architectures; lab-scale cell assembly/testing; select lead architecture • **Months 18–36:** Polymer scale synthesis + R2R film/lamination; confirm performance retention; deliver scale-up protocol

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**A. Technical Leadership and Core Material Innovation** The research institutes (RPI and UIC) serve as the technology originators for this project, with primary responsibility for advancing polymeric electrolyte materials as a core enabling technology for LSBs. Their primary responsibility is to design, synthesize, and validate the advanced polymer materials before they are transferred to the Korean SME for industrial scale-up. **B. Research Scope and Key Technical Contributions** Within the defined research scope, the institute will conduct polymer synthesis and formulation, including the development of QSSE-ready matrices, composite-compatible host polymers, and layered electrolyte components with tailored mechanical and interfacial properties. The RPI will perform comprehensive physicochemical characterization, including thermal, mechanical, morphological, and chemical analyses, to guide material optimization and down-selection. In parallel, UIC will lead electrochemical characterization at the materials level, including ionic conductivity, Li^+ transference number, electrochemical stability window, polysulfide crossover behavior, and Li metal interfacial impedance evolution. These data will serve as benchmarks for evaluating performance improvements relative to conventional polymer electrolytes. **C. Laboratory-Scale Architecture Demonstration and Cell Testing** The RPI & UIC will design and demonstrate quasi-solid-state, composite, and multilayer polymer electrolyte architectures and integrate them into laboratory-scale LSB cells. This includes electrolyte film preparation, electrode–electrolyte interface engineering, cell assembly, and electrochemical testing under practical conditions. The RPI & UIC will diagnose performance-limiting mechanisms such as polysulfide shuttle, interfacial degradation, and mechanical failure, and will iterate materials design accordingly. **D. Knowledge Transfer and Collaboration** The RPI & UIC will play a key role in knowledge generation and transfer to the Korean SME partner. This includes sharing polymer formulations, processing conditions, characterization protocols, and performance data to support polymer scale-up and roll-to-roll manufacturing efforts.
Expected R&R of Korean SME Partner	**A. Commercial-Level Cell Assembly and System Integration** The Korean SME partner will serve as the industrial cell integration and commercialization lead, with primary responsibility for translating polymer electrolyte materials supplied by RPI into commercially relevant LSB cells. The SME will focus on cell design, assembly, and process integration, ensuring that the developed polymer electrolytes are compatible with practical electrode fabrication, stacking/lamination, and cell packaging methods consistent with industrial battery manufacturing practices. **B. Prototype and Commercial-Format Cell Development** The SME partner will assemble laboratory-to-commercially relevant LSB cell formats (e.g., advanced coin, pouch, or prototype cells as appropriate), incorporating the project-developed polymer electrolyte materials into quasi-solid-state, composite, or multilayer configurations. Emphasis will be placed on manufacturability, mechanical robustness, electrode–electrolyte contact quality, and process repeatability, rather than materials synthesis. The SME will adapt cell assembly parameters—including electrode loading, electrolyte integration, and stack pressure—to maximize performance under practical operating conditions. **C. Testing, Validation, and Performance Verification** A key responsibility of the Korean SME will be performance validation at the cell and system level, using testing protocols aligned with commercial battery evaluation. This includes cycling stability, Coulombic efficiency, rate capability, and preliminary safety and reliability testing relevant to LSB commercialization. The SME will generate performance and durability data needed to assess readiness for downstream pilot demonstration and customer engagement. **D. Commercialization Preparation and Market Readiness** The SME partner will lead commercialization planning activities, including assessment of manufacturability, cost structure, supply-chain compatibility, and target application requirements. The SME will evaluate pathways for transitioning project outcomes into commercial LSB products,

	identify potential end users and markets, and define next-step requirements for pilot production and regulatory or certification considerations. **E. Collaboration and Technology Transfer** The SME will work closely with RPI & UIC through regular technical coordination, data exchange, and joint troubleshooting to ensure effective technology transfer at the cell integration level. Feedback from commercial-format cell testing will be communicated to RPI to inform further material optimization and performance refinement.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**A. Technological Achievements: A Breakthrough in "Post-Li-ion" Storage** This joint R&D project is expected to deliver next-generation polymer electrolyte materials that address the fundamental limitations of existing polymeric and inorganic solid electrolytes for LSBs. Key technological outcomes include the development of scalable polymer electrolyte matrices with improved Li^+ conductivity/selectivity, effective polysulfide suppression, strong interfacial adhesion, and robust electrochemical and mechanical stability. These materials will be demonstrated across quasi-solid-state, composite, and multilayer electrolyte architectures. At the cell level, the project will demonstrate laboratory-scale LSBs incorporating the developed polymer electrolytes under practical conditions (lean electrolyte, meaningful sulfur loading), validating improved cycling stability, Coulombic efficiency, and interfacial durability relative to conventional liquid and polymer electrolyte systems. The ability to integrate inorganic ion conductors and layered designs without sacrificing processability or interface stability will represent a significant advance over current solid-state and hybrid electrolyte technologies. **B. Commercialization Outcomes** The project is explicitly structured to enable technology transfer and commercialization readiness. Polymer electrolyte materials developed and scaled by RPI will be transferred to the Korean SME partner for commercial-level LSB cell assembly, testing, and validation. This collaboration will establish a clear pathway from materials innovation to industry-relevant battery prototypes, reducing the time and risk associated with commercialization. Expected commercialization outcomes include: (1) Polymer electrolyte materials compatible with existing battery manufacturing workflows; (2) Demonstration of commercial-format LSB cells assembled and tested by the Korean SME; (3) Defined performance benchmarks and manufacturing requirements for pilot-scale production; and (4) Strengthened capability of the Korean SME to internalize advanced overseas materials technology and prepare for market entry **C. Industrial and Economic Impact** • **Reduction of Critical Mineral Dependence**: The LSB chemistry eliminates the need for Cobalt and Nickel, which are subject to volatile pricing and geopolitical supply constraints. This shift enhances national energy security for both Korea and the US. • **Support for Carbon Neutrality**: By enabling Long-Duration Energy Storage (10+ hours), this technology directly facilitates the integration of intermittent renewable energy (wind, solar) into the grid, helping to replace fossil-fuel peaker plants. A 100 MWh deployment is estimated to avoid up to 18,000 tons of CO_2 emissions. • **Expansion of the Solid-State Ecosystem:** The success of this "Polymer-Based" approach will stimulate the broader solid-state battery industry, proving that polymer electrolytes are a viable, processable alternative to brittle ceramic electrolytes for large-format applications. • **Workforce Development**: This joint R&D project will strengthen workforce development by building practical expertise in advanced polymer electrolyte materials, battery integration, and commercial-level cell validation within the Korean SME. Hands-on involvement in cell assembly, testing, and manufacturability assessment will cultivate industry-ready engineers and support job creation as the SME expands its technical workforce. By internalizing advanced materials and manufacturing know-how, the project will accelerate SME growth and competitiveness in next-generation secondary battery markets.

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6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization